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Course: National Higher Diploma in Mechanical Engineering
Module Name: Solid Mechanics IV
Module Code: SME3701
Assignment: 1 (One)

Question 1 **[22]**

Consider a single-degree-of-freedom (SDOF) mass–spring system undergoing undamped free vibration, as illustrated in Figure 1. The system consists of a mass of 5 kg attached to a spring with stiffness 2000 N/m. The system is initially displaced by 0.02 m and released from rest with zero initial velocity. The motion of the system is described by the equation $x(t)$= $x0\cos(\omega nt)$, where ωn is the natural angular frequency.

1.1 Determine the natural angular frequency ωn of the system using MATLAB. (3)
1.2 Develop a MATLAB script to compute the displacement $x(t)$ of the system. (2)
1.3 Plot the displacement versus time for the interval $0 \leq t \leq 5$ seconds. (7)
1.4 Determine the velocity $v(t)$ using MATLAB and plot the velocity versus time on a separate graph. (7)
1.5 Comment on the nature of the motion, the effect of zero damping on the system response, and the relationship between displacement and velocity. (3)

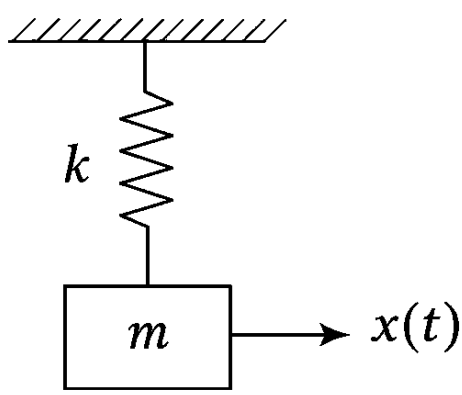


Figure 1: Undamped Free Vibration of a Mass–Spring System.

Given Data: $m = 5\ kg, \quad k = 2000\ N/m, \quad x_0 = 0{,}02\ m, \ and \quad v_0 = 0\ m/s$

1.1 Determine the natural angular frequency ωn of the system using MATLAB. (3)

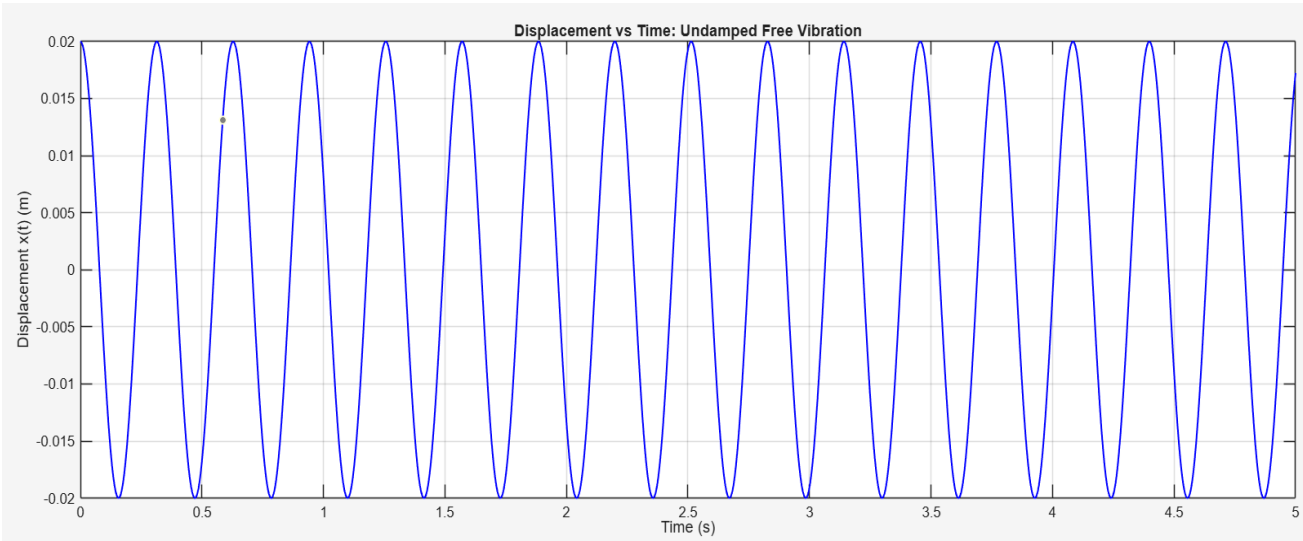
$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{2000}{5}} = 20\ rad/s$$

The MATLAB results should confirm the 20.0000 value for natural angular frequency.

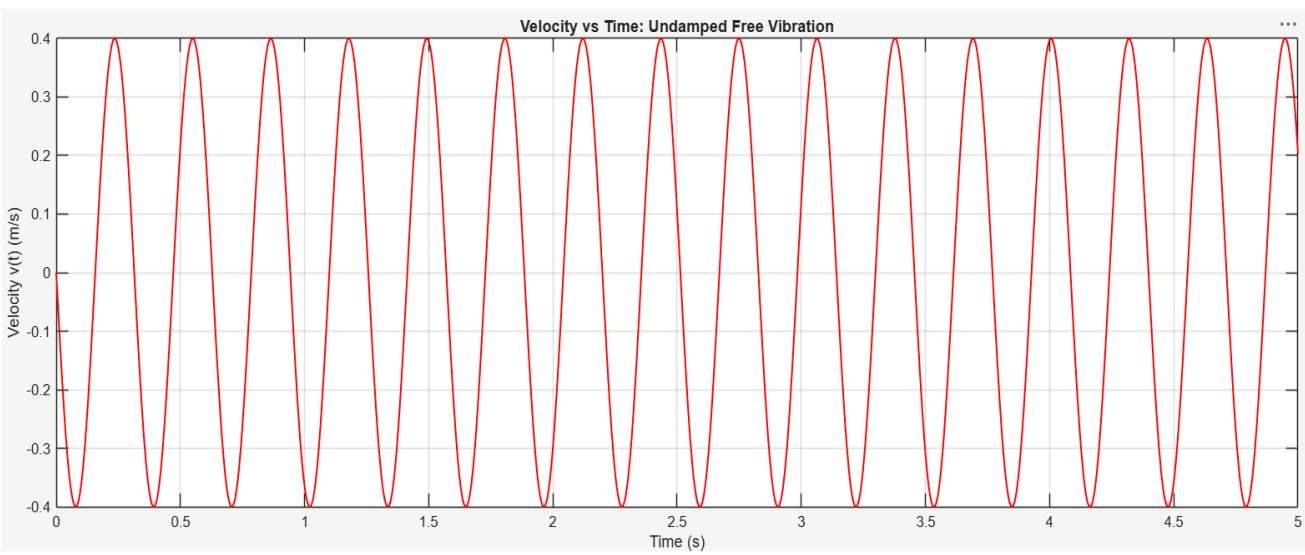
1.2 Develop a MATLAB script to compute the displacement $x(t)$ of the system. (2)

```
% SDOF undamped free vibration: mass-spring system
m = 5;                          % mass (kg)
k = 2000;                       % stiffness (N/m)
x0 = 0.02;                      % initial displacement (m)
v0 = 0;                         % initial velocity (m/s)

wn = sqrt(k/m);                 % natural angular frequency (rad/s)

t = linspace(0,5,1000);         % time vector from 0 to 5 s
x = x0 * cos(wn * t);           % displacement time history
v = -x0 * wn * sin(wn * t);     % velocity time history (derivative of x)

% Display natural frequency
fprintf('Natural angular frequency wn = %.4f rad/s\n', wn);

% Plot displacement
figure;
plot(t, x, 'b', 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Displacement x(t) (m)');
title('Displacement vs Time: Undamped Free Vibration');
grid on;

% Plot velocity
figure;
plot(t, v, 'r', 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Velocity v(t) (m/s)');
title('Velocity vs Time: Undamped Free Vibration');
grid on;
```

1.3 Plot the displacement versus time for the interval $0 \leq t \leq 5$ seconds. (7)



1.4 Determine the velocity $v(t)$ using MATLAB and plot the velocity versus time on a separate graph. (7)



1.5 Comment on the nature of the motion, the effect of zero damping on the system response, and the relationship between displacement and velocity. (3)

Nature of the Motion:

- The motion is Simple Harmonic Motion (SHM).
- Because the system is undamped and linear, the displacement follows a perfect sinusoidal pattern (cosine in this case).

Effect of Zero Damping:

- This problem specifies zero damping.
- As a result, there is no energy dissipation mechanism.
- The total mechanical energy (sum of kinetic and potential energy) remains constant.

Relationship between Displacement and Velocity:

- The displacement and velocity are not in phase.
- When the displacement is at its maximum (amplitude), the velocity is zero.
- When the displacement is zero (passing through the equilibrium point), the velocity is at its maximum magnitude.
- Velocity is the time derivative of displacement. (As seen in the equations).

GitHub Repository Link

https://github.com/motsekinoaharamasilo/SME3701-Assignment1.git

TOTAL MARKS: 22

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